fusion product, the secondary nucleus and forms the **primary endosperm nucleus (PEN)**. Since this involves the fusion of three nuclei, this phenomenon is called **triple fusion**. This act results in endosperm formation which forms the nutritive tissue for the embryo.

1.6 Post Fertilization structure and events

After fertilization, several changes take place in the floral parts up to the formation of the seed (Figure 1.20).

The events after fertilization (endosperm, embryo development, formation of seed, fruits) are called post fertilization changes.

Parts before fertilization	Transformation after fertilization
Sepals, petals, stamens, style and stigma	Usually wither and fall off
Ovary	Fruit
Ovule	Seed
Egg	Zygote
Funicle	Stalk of the seed
Micropyle (ovule)	Micropyle of the seed(facilitates O_2 and water uptake)
Nucellus	Perisperm
Outer integument of ovule	Testa (outer seed coat)
Inner integument	Tegmen (inner seed coat)
Synergid cells	Degenerate
Secondary nucleus	Endosperm
Antipodal cells	Degenerate

Endosperm

The primary endosperm nucleus (PEN) divides immediately after fertilization but before the zygote starts to divide, to form the endosperm. The primary endosperm nucleus is the result of triple fusion (two polar nuclei and one sperm nucleus) and thus has 3n number of chromosomes. It is a nutritive tissue and regulatory structure that nourishes the developing embryo. Depending upon the mode of development three types of endosperm are recognized in angiosperms. They are nuclear endosperm, cellular endosperm and helobial endosperm (Figure 1.21).

Nuclear endosperm: Primary Endosperm Nucleus undergoes several mitotic divisions without cell wall formation thus a free nuclear condition exists in the endosperm. Examples: *Coccinia*, *Capsella* and *Arachis*

Cellular endosperm: Primary endosperm nucleus divides into 2 nuclei and it is immediately followed by wall formation. Subsequent divisions also follow cell wall formation. Examples: *Adoxa, Helianthus* and *Scoparia*

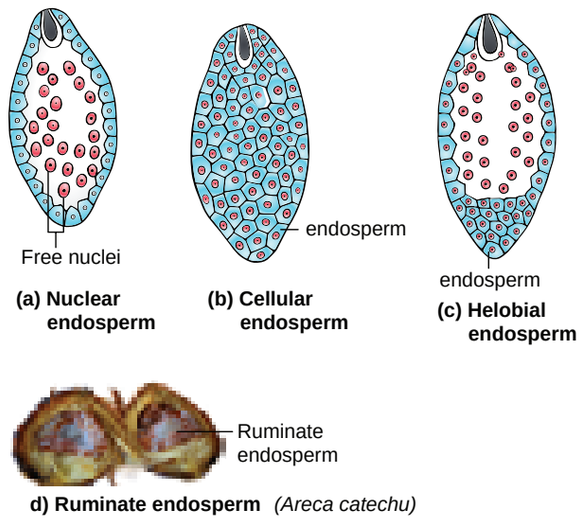


Figure 1.21 Types of Endosperm

Helobial endosperm: Primary Endosperm Nucleus moves towards base of embryo sac and divides into two nuclei. Cell wall formation takes place leading to the formation of a large micropylar and small chalazal chamber. The nucleus of the micropylar chamber undergoes several free nuclear division whereas that of chalazal chamber may or may not divide. Examples : *Hydrilla* and *Vallisneria*.

The endosperms may either be completely consumed by the developing embryo or it may persist in the mature seeds. Those seeds without endosperms are called non- endospermous or ex- albuminous seeds. Examples: Pea, Groundnut

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